

Trustworthy 3D Perception, Calibrated Foundation Models, and Geometric Consensus

Postdoctoral Statement of Research Interests and Vision

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1. Research Vision: Trustworthy Visual AI for Real-World 3D Environments

Next-generation autonomous systems—from planetary rovers and space proximity inspectors in zero gravity to terrestrial industrial robotics—must operate in dynamic, unstructured physical environments. While web-scale visual foundation models exhibit impressive general visual knowledge, deploying them in safety-critical domains poses three foundational challenges:

- Uncalibrated Epistemic Uncertainty:** Deep visual encoders frequently generate overconfident predictions on out-of-distribution observations, lacking calibrated measures of their own contextual reliability.
- Premature Decision Collapse:** Standard perceptual architectures discard secondary semantic hypotheses via greedy argmax selection, eliminating the distributional nuance required for robust multi-sensor arbitration.
- Supervision Scarcity in 3D/Physical Domains:** Dense 3D point-cloud and multi-sensor annotations are prohibitively expensive to acquire, demanding principled, label-free physical consistency objectives.

My doctoral research has addressed the mathematical and systems foundations of these challenges: formulating non-parametric inference-time coordination ($\Delta\theta = 0$), proving the benefits of semantic evidence preservation, and deriving label-free process rewards from multi-view physical consistency. Within the Computer Vision, Machine Intelligence and Imaging (**CVI²**) research group at **SnT**, I propose to harness these foundations to build reliable, calibrated, and self-supervised 3D visual perception systems.

2. Doctoral Methodological Foundations and Empirical Breakthroughs

My doctoral dissertation, *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations* (supervised by Prof. Ebenezer Owusu), contributes three core pillars:

2.1 Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)

In research in preparation for *IEEE TPAMI*, I formulated **RCASA**, a probabilistic framework that coordinates heterogeneous, frozen visual foundation models at inference time. By formally decoupling candidate-level classifier confidence from contextual reliability using Platt scaling and Bernoulli uncertainty estimation, RCASA guarantees class-wise bounded updates and probability simplex preservation. This enables autonomous systems to arbitrate between competing hypotheses non-parametrically, preventing catastrophic errors without continual fine-tuning.

2.2 The Physics of Semantic Retention: Beyond Argmax (ICLR 2027 Submission)

A major discovery of my doctoral work—formalized in our recent submission to **ICLR 2027**, titled *Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation*—is that premature decision collapse severely bottlenecks multi-model composition. Discarding sub-maximal semantic hypotheses at encoder outputs via hard argmax permanently truncates distributional alternatives. In matched causal interventions across 312 scenes (~50M points) of ScanNet200 and ScanNet++, I showed that preserving compact top- k candidate distributions enables downstream geometric consensus to recover overlooked novel categories,

boosting segmentation harmonic mean from 28.47 to 34.87 (+6.40 points).

2.3 Geometric Lifting and Multi-View Physical Agreement (ACCV 2026, SCAIR 2026)

In ACCV 2026 (Paper #482) and SCAIR 2026, I constructed 2D-to-3D projective pipelines connecting camera poses (K, T) with multi-view point-cloud consensus. Furthermore, we demonstrated that physical agreement across viewpoints provides an intrinsic, label-free process reward for training multimodal LLMs (**Qwen2.5-VL-7B** via **GRPO/VERL** on **vLLM**), neutralizing spatial area-bias gaming ($\rho > 0.72 \rightarrow 0.334$) without human-labeled reward models.

3. Proposed Postdoctoral Research Program at CVI² (SnT)

At the University of Luxembourg, I aim to expand these foundations across three high-impact research themes aligned with CVI² and SnT priorities:

Theme A: Calibrated Multi-Sensor 3D Perception and Anomaly Detection

Safety-critical systems rely on heterogeneous sensors (RGB-D, LiDAR, event cameras). I propose to develop a calibrated multi-sensor fusion architecture that continuously quantifies epistemic uncertainty. Using distributional semantic retention and reliability calibration, the framework will dynamically down-weight degraded or sensor-occluded modalities, detecting anomalies and out-of-distribution objects with mathematically certified confidence intervals.

Theme B: Proximity Operations and Pose Estimation in the Zero-G Lab

Autonomous rendezvous and docking in space demand robust 6D pose estimation and shape analysis under severe illumination shifts and uncooperative target geometries. Harnessing CVI²'s specialized **Zero-G Lab**, I propose to develop distribution-preserving geometric tracking pipelines that project multi-camera viewpoints into unified 3D point-cloud representations. By preserving multi-hypothesis distributions through time, the tracker will prevent catastrophic drift during critical proximity operations.

Theme C: Self-Supervised Foundation-Model Adaptation via Geometric Invariance

To eliminate the requirement for massive ground-truth 3D annotations in novel domains, I propose to extend my label-free process-reward framework to visual generative models and 3D foundation models. By enforcing epipolar constraints, photometric reprojection consistency, and multi-view geometric consensus as self-supervision, models can adapt autonomously to new environments using high-performance GPU resources (LuxProvide MeluXina supercomputer).

4. Synergies with CVI², SnT, and Long-Term Impact

The CVI² group led by Prof. Djamila Aouada is globally renowned for its high-impact research in 3D computer vision, visual generative modeling, and multi-sensor intelligence. SnT's unique ecosystem provides an ideal environment for translating theoretical reliability guarantees into deployable space and industrial technologies.

My background in PyTorch, 3D point-cloud geometry, CUDA-accelerated streaming inference, and rigorous statistical methodology equips me to contribute immediately to CVI²'s project portfolio, collaborate with PhD researchers, and produce top-tier publications at CVPR, ICCV, ECCV, ICLR, and IEEE TPAMI.